

Demonstrating Controller-Free Mixed Reality X-Ray Machine Training

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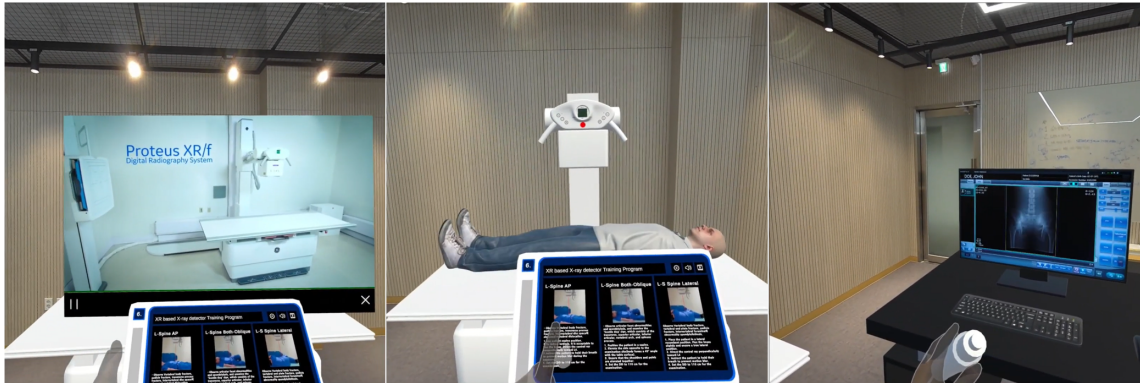


Figure 1: Teaser image illustrating the end-to-end training workflow of the proposed controller-free Mixed Reality system using entirely virtual objects. The left panel shows a virtual representation of an X-ray setup in a real-world spatial context, the center panel depicts hands-on operation of a virtual X-ray machine and patient using natural hand interaction with the Virtual Tablet, and the right panel presents a virtual imaging output interface. The system enables walk-up, immersive training in a live demo setting without physical equipment or controllers.

ABSTRACT

We demonstrate a controller-free, Mixed Reality training system that enables hands-on X-ray machine operation through natural hand interaction. The system addresses key limitations of conventional medical device training, including restricted physical access, high equipment costs, and radiation exposure risks. By leveraging optical hand tracking on a standalone head-mounted display, users can interact with a virtual X-ray machine using gestures that closely resemble real-world operation. A central component of the demo is a Virtual Tablet interface, which provides video-based operational guidance and allows users to seamlessly select and execute training scenarios for different patient postures. All interactions—including device manipulation and user interface navigation—are performed entirely through hand tracking, enabling immediate, walk-up use without controllers or calibration overhead. The system is designed specifically for live demo settings, allowing first-time users to experience realistic medical device operation within minutes. Informal user feedback from nursing students and clinical professionals indicates high engagement, strong perceived realism, and clear potential for scalable deployment in medical education and clinical training environments.

Index Terms: Mixed Reality, Medical Device Training, Hand Tracking, Controller-Free Interaction, X-Ray Machine

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1 INTRODUCTION

The increasing complexity of modern medical equipment has made hands-on, practice-based training essential for healthcare professionals [1, 3]. In particular, X-ray imaging systems require precise spatial understanding and procedural accuracy, yet conventional training methods remain severely constrained by limited physical access, high equipment costs, and the inherent risks associated with radiation exposure [2, 4]. These constraints significantly limit training opportunities, especially for students and novice technologists.

Mixed Reality (MR), as a subset of Extended Reality, has emerged as a promising alternative for medical education by enabling safe, repeatable training environments without physical or radiological constraints. However, many existing MR-based medical training systems rely on handheld controllers, predefined interaction workflows, or desktop-style interfaces. Such designs often reduce realism and make these systems less suitable for immediate, walk-up use in live demo settings, where first-time users must be able to engage with minimal instruction.

In this work, we demonstrate a controller-free, Mixed Reality (MR) training system for X-ray machine operation, designed explicitly for live demo settings and hands-on experience. Leveraging optical hand tracking on a standalone head-mounted display, the system allows users to manipulate virtual medical equipment using natural hand gestures that closely resemble real-world operation. By eliminating physical controllers, the system minimizes setup overhead and enables conference attendees to engage with the training environment immediately.

A central design element of the system is the Virtual Tablet interface, which serves as the primary interaction and learning hub. Inspired by the widespread use of tablets in clinical environments, the Virtual Tablet provides intuitive access to video-based operational guidance and scenario selection for different patient postures. All interactions—including tablet navigation and device manipulation—are performed entirely through hand tracking, reinforcing immersion and interaction continuity.

Unlike prior XR medical training systems that primarily emphasize long-term curriculum integration, this work focuses on experiential clarity, accessibility, and robustness in a demo context. The system is designed to be experienced within minutes, making it well suited for IEEE VR demo sessions where immediate usability and experiential impact are critical. Through this demo, participants can directly experience how controller-free interaction and Mixed Reality can support realistic, safe, and scalable medical device training.

2 DEMO-ENABLING SYSTEM DESIGN

The system is designed to support immediate, walk-up interaction in live demo environments such as IEEE VR demo sessions. Rather than emphasizing internal architecture or implementation details, the design prioritizes minimal setup overhead, interaction robustness, and experiential clarity for first-time users. Two key elements enable this demo-oriented approach: controller-free hand interaction and an intuitive Virtual Tablet interface.

2.1 Controller-Free Hand Interaction

The demo employs optical hand tracking to allow users to interact with the virtual X-ray machine using natural hand gestures, without relying on physical controllers. This design choice removes the need for device familiarization or calibration, enabling conference attendees to engage with the system immediately upon wearing the headset.

Users can perform essential machine operations—such as adjusting the patient table, repositioning the X-ray arm, and activating predefined functions—through direct hand manipulation that closely mirrors real-world operation. The interaction design emphasizes gesture simplicity and tolerance, allowing stable operation despite imprecise movements or brief tracking loss, which are common in public demo settings. This robustness ensures that the system remains responsive and usable across a wide range of user behaviors and environmental conditions.

2.2 Virtual Tablet for Walk-Up Use

To guide users through the demo experience, the system integrates a Virtual Tablet interface that serves as the primary learning and control hub. The tablet metaphor was intentionally chosen due to its familiarity in clinical environments, allowing users to quickly understand its function without external instruction.

The Virtual Tablet provides access to video-based operational guidance and enables users to select training scenarios corresponding to different patient postures. It can be summoned at any time through a simple activation gesture, keeping the environment visually uncluttered while maintaining accessibility. By consolidating instruction, navigation, and scenario control into a single interface, the tablet supports a smooth transition from observation to hands-on interaction within a short demo session.

3 DEMO EXPERIENCE

During the live demo, attendees can immediately engage with the system by summoning the Virtual Tablet and selecting a training scenario. Within a few minutes, users experience hands-on operation of the virtual X-ray machine through controller-free hand interaction, including adjusting device components and observing system responses in Mixed Reality. The demo is designed to be completed within a short session, allowing multiple participants to sequentially experience the system without recalibration or setup overhead.

4 USER EVALUATION SUMMARY

We conducted an informal user study to assess the usability and experiential quality of the system in short, demo-style sessions. A total of 22 participants took part in the study, including 20 undergraduate nursing students from Gangneung–Wonju National University,

one nurse from the Severance Hospital nursing team, and one expert from the functional examination team at Seoul St. Mary's Hospital.

Each participant experienced the system for approximately 10 minutes, reflecting the time constraints and interaction patterns typical of conference demo environments. During the session, participants interacted with the virtual X-ray machine using controller-free hand tracking, accessed instructional content through the Virtual Tablet, and performed basic operational tasks.

Overall feedback was consistently positive. Nursing students reported that the hand-based interaction provided a practice environment closely resembling real-world clinical operation, enhancing immersion and engagement. One student noted that “using hands for interaction creates a realistic practice environment similar to actual clinical training.” Clinical experts emphasized the system's ability to overcome spatial constraints and provide natural medical device interaction without physical equipment, highlighting its effectiveness in conveying operational understanding even within a short demo session.

These results indicate that the system is well suited for walk-up demo scenarios, offering immediate usability, strong perceived realism, and clear educational value for both novice learners and experienced practitioners.

5 CONCLUSION

This demo presented a controller-free, Mixed Reality training system for operating X-ray medical equipment, designed for immediate use in live demo environments. By combining natural hand tracking interaction with an intuitive Virtual Tablet interface, the system enables first-time users to experience realistic medical device operation without setup overhead or physical controllers. While some physical controls—such as foot-pedal interactions—were adapted into hand-based interaction due to implementation constraints within the current development scope, the system remains highly intuitive.

The demo illustrates how Mixed Reality can effectively address spatial and safety constraints inherent in conventional medical training while maintaining a high level of realism and user engagement. Informal feedback from nursing students and clinical professionals confirms that meaningful hands-on experience and operational understanding can be achieved even within short, walk-up demo sessions.

Overall, this work demonstrates the feasibility and value of walk-up, immersive medical device training and highlights the potential of controller-free Mixed Reality as a practical and scalable approach for medical device education and training applications.

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